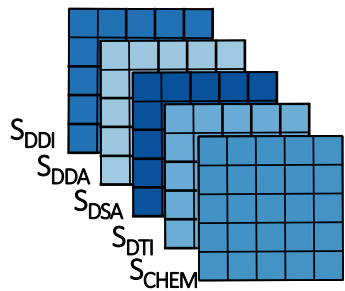
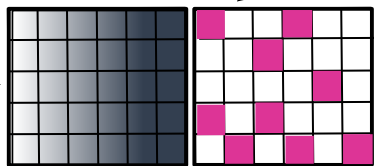


Drug similarity matrices

- Drug-drug interactions (DDI)
- Drug-disease associations (DDA)
- Drug-side effects (DSA)
- Drug-target interactions (DTI)
- Drug chemical



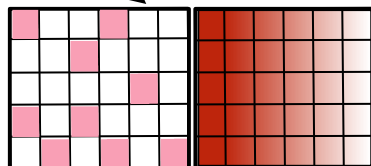
$$H = \frac{\sum_j \beta_j S_j^r}{\sum_j \beta_j}$$



H

X_r

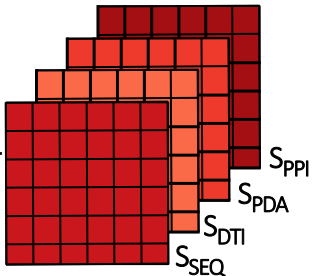
+



X_c

W

$$W = \frac{\sum_j \alpha_j S_j^c}{\sum_j \alpha_j}$$



Target similarity matrices

- Protein-protein interactions (PPI)
- Protein-disease associations (PDA)
- Drug-target interaction (DTI)
- Protein sequence

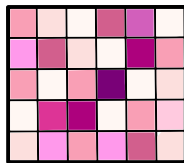
row-wise
normalisation

column-wise
normalisation



X

Drugs



Targets

$$\hat{X} = (1 - \gamma)HX_r + \gamma X_c W$$

Drug self-representation model

Target self-representation model